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EX.NO.: 1	PROBLEM STATEMENT

AIM: To prepare a PROBLEM STATEMENT for any project.

REQUIREMENTS:

Hardware Interfaces

- Pentium(R) 4 CPU 2.26 GHz, 128 MB RAM
- Screen resolution of at least 800 x 600 required for proper and complete viewing of screens. Higher resolution would not be a problem.
- CD ROM Driver

Software Interfaces

- Any Windows-based operating system
- WordPad or Microsoft Word

THEORY:

The problem statement is the initial starting point for a project. It is basically a one to three page statement that everyone on the project agrees with that describes what will be done at a high level. The problem statement is intended for a broad audience and should be written in non-technical terms. It helps the non-technical and technical personnel communicate by providing a description of a problem. It doesn't describe the solution to the problem.

The input to requirement engineering is the problem statement prepared by customer. It may give an overview of the existing system along with broad expectations from the new system.

The first phase of requirements engineering begins with requirements elicitation i.e. gathering of information about requirements. Here, requirements are identified with the help of customer and existing system processes. So from here begins the preparation of problem statement.

So, basically a problem statement describes **what** needs to be done without describing **how**.

Conclusion:

The problem statement was written successfully by following the steps described above.

EX.NO.: 2	SYSTEM REQUIREMENTS SPECIFICATION

AIM: Understanding an SRS.

REQUIREMENTS:

Hardware Requirements:

- PC with 300 megahertz or higher processor clock speed recommended; 233 MHz minimum required.
- 128 megabytes (MB) of RAM or higher recommended (64 MB minimum supported)
- 1.5 gigabytes (GB) of available hard disk space
- CD ROM or DVD Drive
- Keyboard and Mouse(compatible pointing device).

Software Requirements:

- Rational Rose, Windows XP,

THEORY:

An SRS is basically an organization's understanding (in writing) of a customer or potential client's system requirements and dependencies at a particular point in time (usually) prior to any actual design or development work. It's a two-way insurance policy that assures that both the client and the organization understand the other's requirements from that perspective at a given point in time.

The SRS document itself states in precise and explicit language those functions and capabilities a software system (i.e., a software application, an e-commerce web site, and so on) must provide, as well as states any required constraints by which the system must abide. The SRS also functions as a blueprint for completing a project with as little cost growth as possible. The SRS is often referred to as the "parent" document because all subsequent project management documents, such as design specifications, statements of work, software architecture specifications, testing and validation plans, and documentation plans, are related to it.

It's important to note that an SRS contains functional and nonfunctional requirements only; it doesn't offer design suggestions, possible solutions to technology or business issues, or any other information other than what the development team understands the customer's system requirements to be.

Characteristics of a good SRS

An SRS should be

a) Correct
b) Unambiguous
c) Complete
d) Consistent
e) Ranked for importance and/or stability
f) Verifiable
g) Modifiable
h) Traceable

Conclusion:

The SRS was made successfully by following the steps described above.

EX.NO.: 3	LIBRARY MANAGEMENT SYSTEM

PROBLEM DEFINITION

The library management system is software, which automates the job of a librarian. The user can inquire about the availability of a book in which he can search by entering the author's name or by entering the title of the book. By confirming the authenticity of a user, the library management system provides information about the number of books already borrowed by the user and by referring to the database whether the user can borrow books or not.

SYSTEM REQUIREMENT SPECIFICATION:

Purpose

The purpose of this SRS is to describe the requirements involved in developing a Library management system.

The intended audience is any person, who wants to inquire, borrow and return the books.

Scope

The product is titled Library Management System. The product will perform the following tasks

- Enquire about the availability of books.
- Borrow books if available.
- Return the borrowed books.

THE OVERALL DESCRIPTION

Hardware interfaces

Hard disk: The database connectivity requires a hardware configuration that is on-line. This makes it necessary to have a fast database system running on high rpm hard disk permitting complete data redundancy and back-up systems to support the primary goal of reliability.

The system must interface with the standard output devise, keyboard and mouse to interact with this software.

Software interfaces

Back End: MS-Access 2007

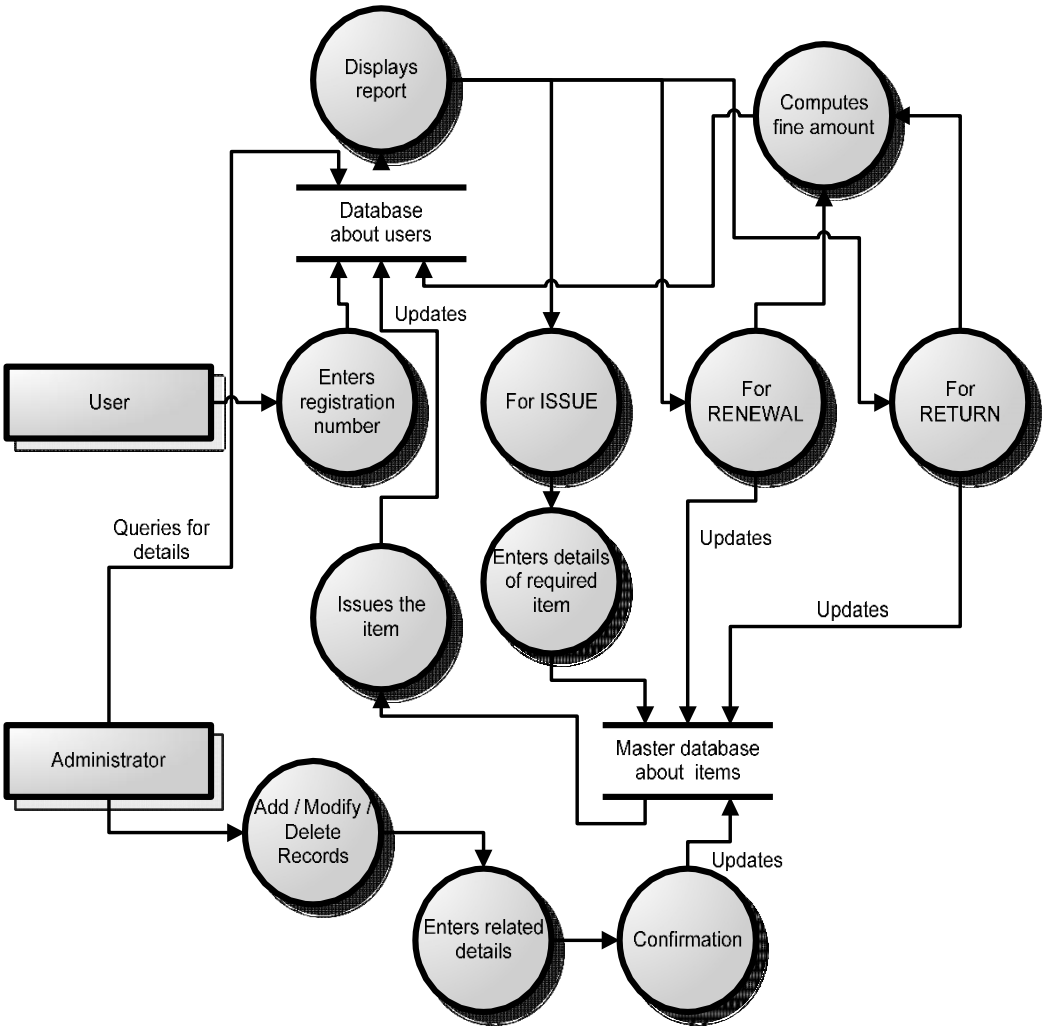
Front End: Microsoft Visual Basic 6.0

DATA STRUCTURES

BOOK DETAILS

FIELD NAME	TYPE	CONSTRAINTS
REGISTER_NO	NUMBER	NOT NULL
BOOK_ID	NUMBER	NOT NULL
ISSUE_DATE	DATE/TIME	
RETURN_DATE	DATE/TIME	
BOOK_NAME	TEXT	

DATA FLOW DIAGRAM



TESTING:

FORM NAME	INPUT	EXPECTED OUTPUT	ACTUAL OUTPUT	STATUS
MAIN MENU FORM	Menu Option	Show required Form	Required Form was displayed	Pass
MEMBERSHIP FORM	Member details are entered	Create new member account	New Account was created	Pass
LOGIN FORM	Member ID and password	If password is correct, login.	Member authenticated for future operations.	Pass
ISSUE FORM	Book ID	If books issued is less than three, issue the book.	Book issued	Pass
RETURN/REISSUE FORM	Book ID	Book returned/ reissued	Book returned/ reissued	Pass
BOOK ENQUIRY	Book Name	Book details are displayed	Book details are displayed	Pass

Conclusion:

Thus the online Library System was implemented using the specified front end and back end tools.

EX.NO.: 4	AIRLINE RESERVATION SYSTEM

PROBLEM DEFINITION

Ticket reservation system for airlines has to be developed. The system developed should contain the following features

1. The system should contain the following features
2. Search for information about the flight by means of flight number and destination
3. While displaying information about the flight it has to provide availability of seats.
4. While reserving tickets the system obtain following information from the user
5. Based on the availability of tickets, the ticket has to be issued. The ticket issued should contain the following information –ticket number, flight no, flight name, date of journey, number of passengers, sex, age and departure time.
6. Cancellation of booked tickets should be available.

SRS DOCUMENT FOR AIRLINE RESERVATION SYSTEM

INTRODUCTION

Purpose

The purpose of this SRS is to describe the requirements involved in developing a Airline Reservation system (ARS) The intended audience is any person who wants to reserve or cancel tickets or to check the availability of Airline tickets

Scope

The product is titled Airline Reservation system (ARS).

The product will perform the following tasks

The software that is being developed can be used to check the availability of the flight tickets for the specified flight, destination and date of journey

If the tickets are available to the users needs and specification, then the software provide a facility to book the tickets.

If the passengers wants to cancel the tickets, he can use the cancellation module of the Airline Reservation System.

THE OVERALL DESCRIPTION

Hardware interfaces

Hard disk: The database connectivity requires a hardware configuration with a fast database system running on high rpm hard-disk permitting complete data redundancy and back-up systems to support the primary goal of reliability.The system must interface with the standard output device, keyboard and mouse to interact with this software.

Software interfaces

Back End: MS Access 2007

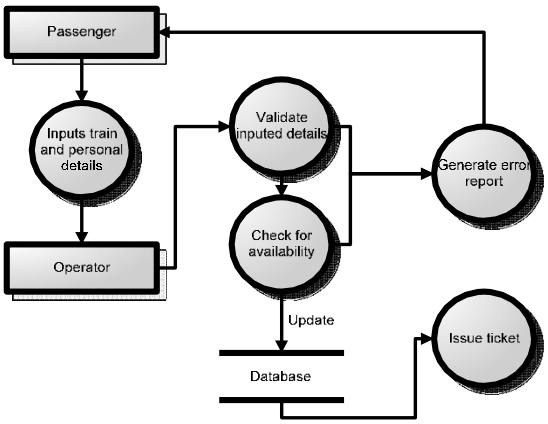
Front End: Microsoft Visual Basic 6.0

DATA STRUCTURES

FLIGHT DETAILS

FIELD NAME	TYPE	CONSTRAINTS
ROUTE_NAME	TEXT	NOT NULL
FLIGHT_NO	NUMBER	NOT NULL
SEATS_AVAIL	NUMBER	
JOURNEY_DATE	DATE/TIME	
DEP_TIME	DATE/TIME	
ARR_TIME	DATE/TIME	
COST	NUMBER	

DATA FLOW DIAGRAM



3.0 TESTING:

FORM NAME	INPUT	EXPECTED OUTPUT	ACTUAL OUTPUT	STATUS
MAIN MENU FORM	Menu Option	Required Form must be displayed	Required form was displayed.	Pass
TICKET AVAILABILITY FORM	Flight route or Flight name	Flight seats availability must be displayed	Flight seats availability are displayed.	Pass
RESERVATION FORM	Personal details were entered.	Ticket must be booked and database updated.	Ticket was booked and database was updated.	Pass
CANCELLATION FORM	Ticket number was entered.	Ticket must be cancelled and database must be updated.	Ticket was cancelled and database was updated..	Pass

Conclusion:

Thus the Airline Reservation System was implemented using the specified front end and back end tools.

EX.NO.: 5	HOTEL MANAGEMENT

PROBLEM DEFINITION

This hotel management system is purposed to make online booking system from a manual data entry booking system. This project is fully developed by Microsoft Access. The online booking system of this hotel makes the customer book the rooms via internet any time and the system automatically store the data inside the system which avoids the error while entering and saving the data and helps to easily modify the data.
This system is developed from the initial stage of system analysis and it covers all the problems definition of the old system, new proposed system and the fact finding reports. Then the system is designed. After the system design the tables are constructed. Creating the database and the tables the system is tested, debugged and implemented.

SYSTEM REQUIREMENT SPECIFICATION:

Purpose

The purpose of this SRS is to describe the requirements involved in developing a Hotel management system.

The intended audience is any person, who wants to book rooms for a specific date without any hassle, online with all the filters available

The product is titled Hotel Management System. The product will perform the following tasks :-

Check the availability of rooms according to the date.

Book rooms online without any difficulty with available online payment options.

If any problem occurs customer can cancel the available rooms according to their need.

THE OVERALL DESCRIPTION

Hardware interfaces

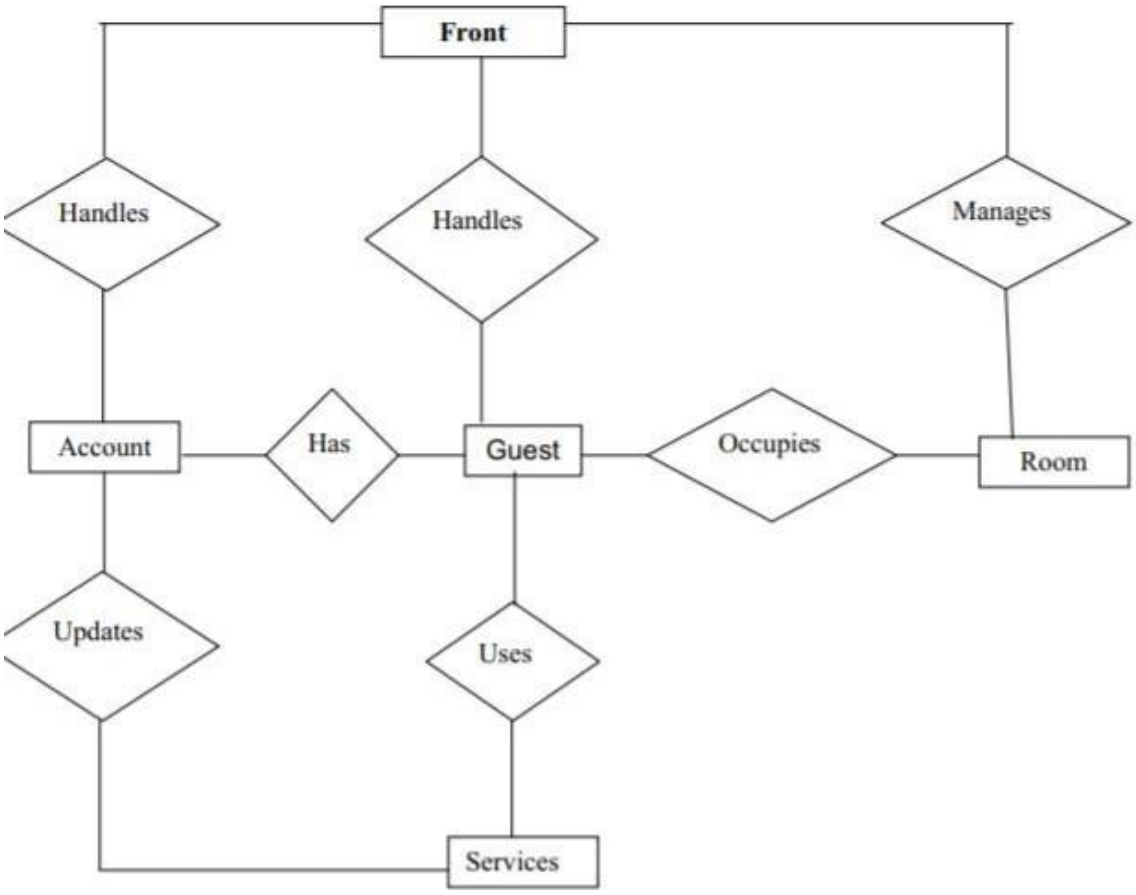
Hard disk: The database connectivity requires a hardware configuration that is online. This makes it necessary to have a fast database system running on high speed hard disk permitting complete data redundancy and back-up systems to support the primary goal of reliability.

The system must interface with the standard output devise, keyboard and mouse to interact with this software.

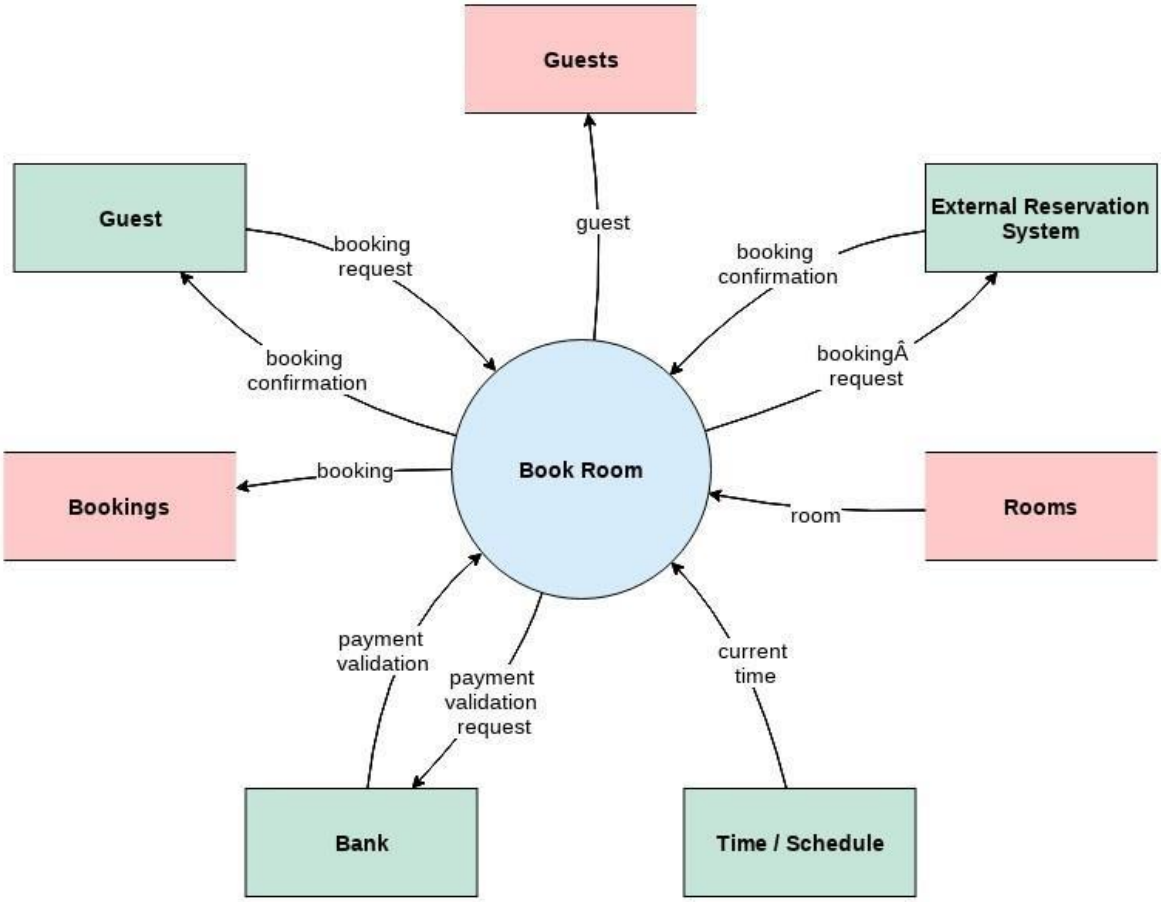
Software interfaces

Back End: MS-Access 2007
Front End: Microsoft Visual Basic 6.0

ER-DIAGRAM



DATA FLOW DIAGRAM



TESTING:

FORM NAME	INPUT	EXPECTED OUTPUT	ACTUAL OUTPUT	STATUS
RESERVATION FORM	Booking Option	Successful registration	Reserved according to availability	Pass
GUEST REGISTRATION	Details enter	Create new member account	New Account was created	Pass

ROOM INFO	rooms availability check	Showing availibity of rooms	shown	Pass
MESSAGE	Message show	Show updated messaged	shown	Pass

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Conclusion:

Thus the online Hotel System was implemented using the specified front end and back end tools.

EX.NO.: 6	ONLINE EXAM REGISTRATION SYSTEM

PROBLEM DEFINITION

Online Exam Registration system.is used in the effective dispatch of registration form to all of the students. This system adopts a comprehensive approach to minimize the manual work and schedule resources, time in a cogent manner. The core of the system is to get the online registration form (with details such as name, reg.no etc.,) filled by the student whose testament is verified for its genuineness by the Exam Registration System with respect to the already existing information in the database. This forms the first and foremost step in the processing of exam application. After the first round of verification done by the system, the information is in turn forwarded to the Exam Controller. The application is then processed manually based on the report given by the system. The system also provides the student the list of exam dates.

SYSTEM REQUIREMENT SPECIFICATION:

Purpose

If the entire process of 'Issue of Hall ticket' is done in a manual manner then it would take several days for the hall ticket to reach the student. Considering the fact that the number of students for hall ticket is increasing every year, an Automated System becomes essential to meet the demand. So, this system uses several programming and database techniques to elucidate the work involved in this process. As this is a matter of National Security, the system has been carefully verified and validated in order to satisfy it.

Scope

The product will perform the following tasks

Students will come to know their status of application and the date in which they must subject themselves for manual document verification.

The System provides an online interface to the user where they can fill in their personal details and submit the necessary documents (may be by scanning).

The controller concerned with the issue of hall ticket can use this system to reduce his workload and process the application in a speedy manner.

Provide a communication platform between the student and the controller.

THE OVERALL DESCRIPTION

Hardware interfaces

The server is directly connected to the client systems. The client systems have access to the database in the server.

Software interfaces

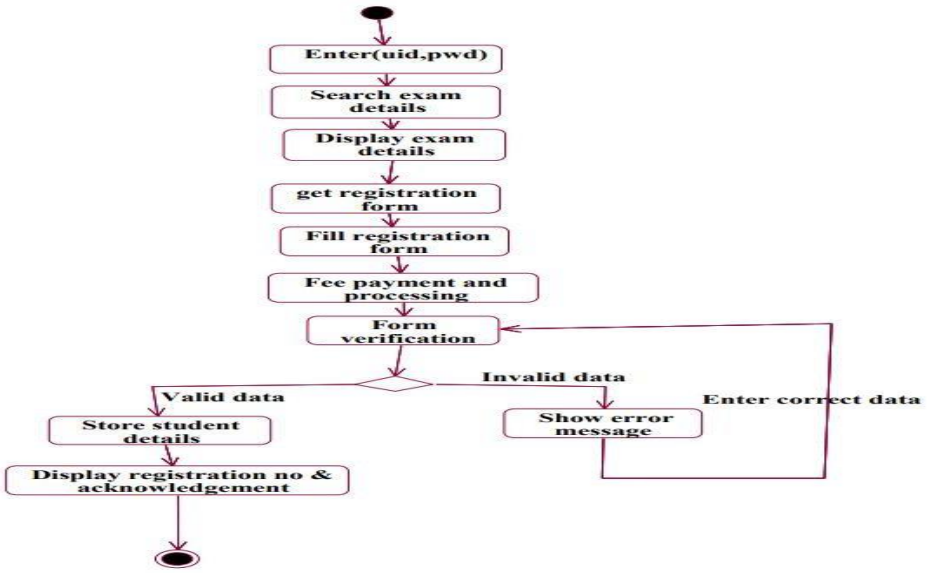
Front End Client - The exporter online interface is built using JSP and HTML.
Web Server – Apache Tomcat Server (Oracle Corporation)
Back End - Oracle 11g database

DATA STRUCTURES

STUDENT DETAILS

FIELD NAME	TYPE	CONSTRAINTS
ST_ID	NUMBER	NOT NULL
ST_NAME	TEXT	NOT NULL
DOB	DATE/TIME	NOT NULL
GENDER	TEXT	NOT NULL
QUALIFICATION	TEXT	NOT NULL
ADDRESS	TEXT	
MOBILE_NO	NUMBER	
EMAIL_ID	TEXT	NOT NULL
USERNAME	TEXT	NOT NULL
PASSWORD	TEXT	NOT NULL

ACTIVITY DIAGRAM:



TESTING:

FORM NAME	INPUT	EXPECTED OUTPUT	ACTUAL OUTPUT	STATUS
COURSE ENROLL PAGE	Hall ticket href	Download the hall ticket	Hall ticket was Downloaded	Pass
REGISTRATION PAGE	Register no, Course Id, Payment ref	If Payment reference was verified, the register the Student.	Authenticated Created for future operations.	Pass
STUDENT ENROLLMENT SEARCH PAGE	Exam Id, Course Id	Search the enrollment data in accordance with exam id or course id.	Returned the Search Result	Pass
ADD COURESE PAGE	Course Id, Course Name, Examiner ID, Examiner Name	Should create a course.	New Course Created	Pass

Conclusion:

Thus, the project for Online Exam Registration system has been successfully executed and codes are generated.

EX.NO.: 7	PASSPORT AUTOMATION SYSTEM

PROBLEM STATEMENT

Passport Automation System is used in the effective dispatch of passport to all of the applicants. This system adopts a comprehensive approach to minimize the manual work and schedule resources, time in a cogent manner. The core of the system is to get the online registration form (with details such as name, address etc.,) filled by the applicant whose testament is verified for its genuineness by the Passport Automation System with respect to the already existing information in the database

SOFTWARE REQUIREMENT SPECIFICATION

PURPOSE

If the entire process of 'Issue of Passport' is done in a manual manner then it would takes several months for the passport to reach the applicant. Considering the fact that the number of applicants for passport is increasing every year, an Automated System becomes essential to meet the demand. So this system uses several programming and database techniques to clarify the work involved in this process. As this is a matter of National Security, the system has been carefully verified and validated in order to satisfy it.

SCOPE

The System provides an online interface to the user where they can fill in their personal details and submit the necessary documents (may be by scanning).

The authority concerned with the issue of passport can use this system to reduce his workload and process the application in a speedy manner.

Provide a communication platform between the applicant and the administrator.

Transfer of data between the Passport Issuing Authority and the Local Police for verification of applicant's information.

Users/Applicants will come to know their status of application and the date in which they must subject themselves for manual document verification.

OVERVIEW DESCRIPTION

HARDWARE INTERFACE

The server is directly connected to the client systems. The client systems have access to the database in the server.

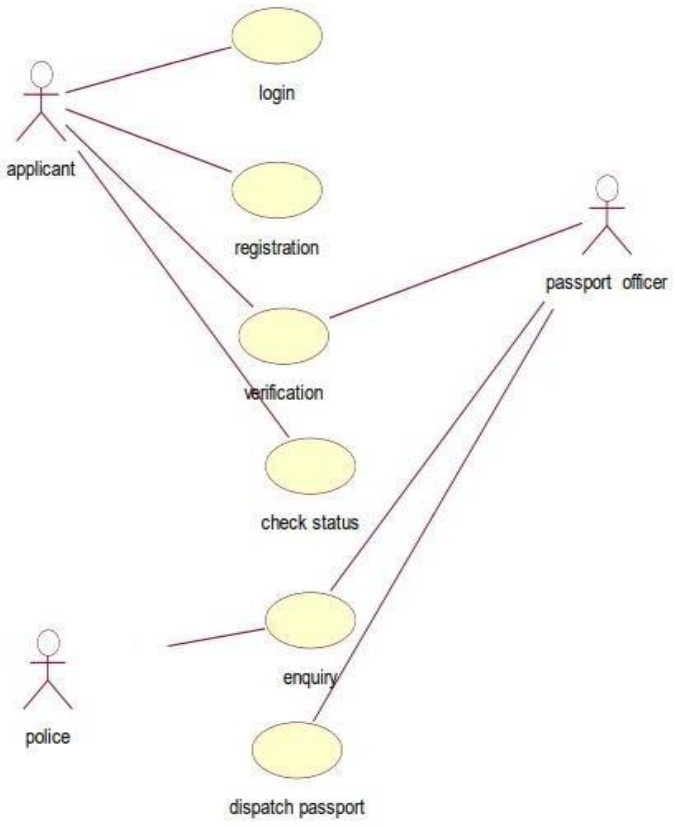
SOFTWARE INTERFACE

Front End Client - The applicant and Administrator online interface is built using JSP and HTML. The Administrator's local interface is built using Java.

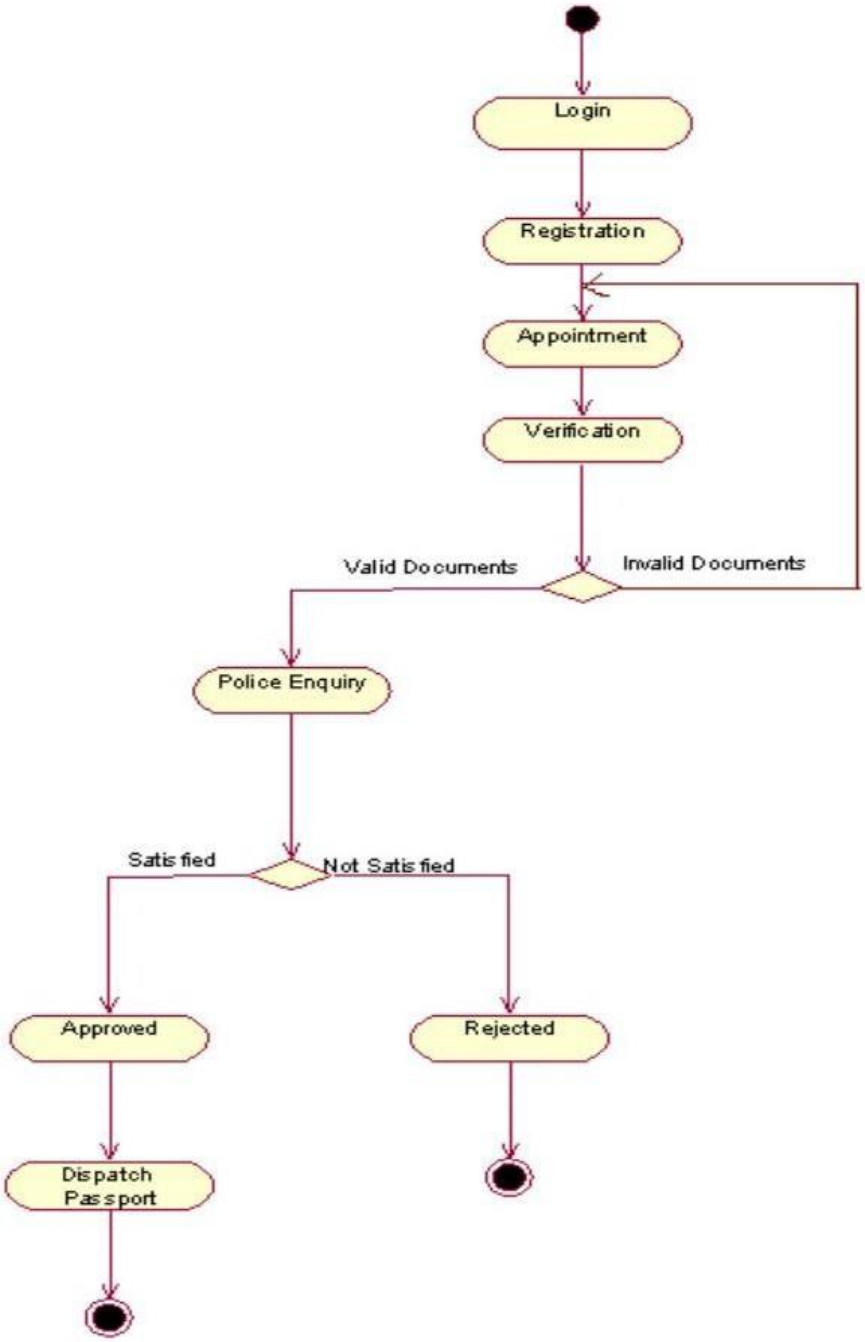
Web Server – Apache Tomcat application server (Oracle Corporation).

Back End – Oracle11g database.

USER CASE DIAGRAM

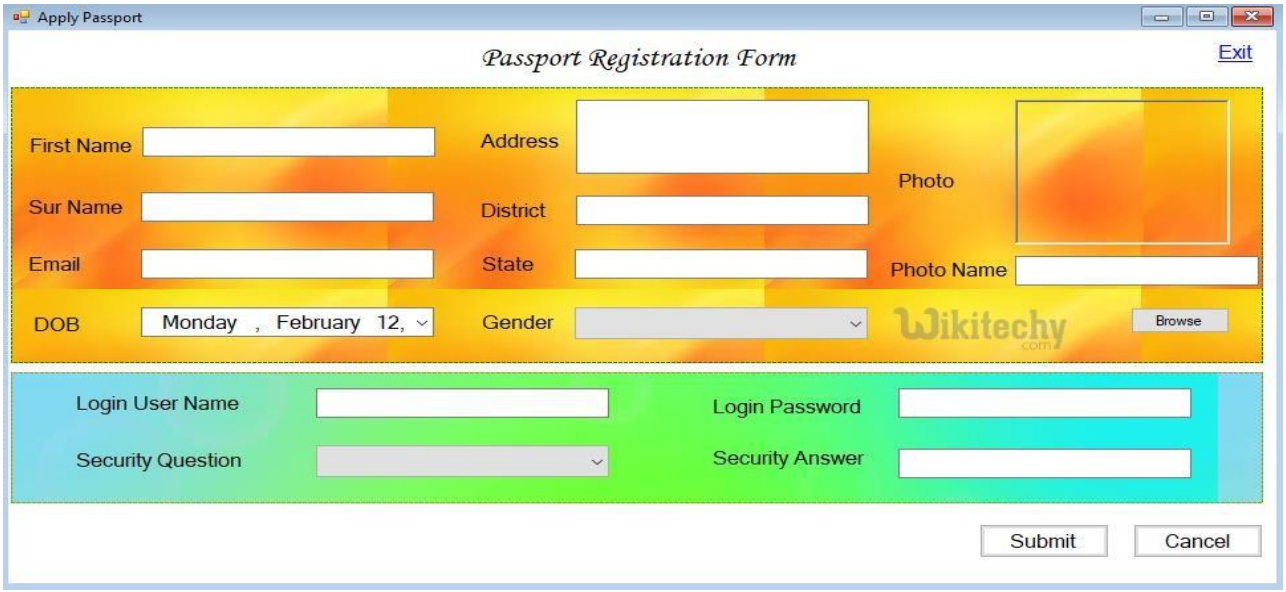


ACTIVITY DIAGRAM



SCREEN SHOTS

New user login



Passport registration form

RESULT

Thus the mini project for passport automation system has been successfully executed and codes are generated.

EX.NO.: 8	VEHICLE PARKING ALLOTMENT SYSTEM

PROBLEM DEFINITION

In this project, we mainly focus on designing a new smart parking system that assists drivers to find parking spaces in a specific parking district. In addition, one of the important aims is to control the traffic created while searching space for parking, this will reduce the air pollution as well as energy consumptions. The main motivation for developing this project is to reduce the traffic congestion which occurs in the urban areas, which is due to the vehicles searching for parking. In newspapers, we are able to see many articles regarding the parking problem all over India in many metropolitan cities like Delhi, Mumbai, Chennai, and Bangalore.

SOFTWARE REQUIREMENT SPECIFICATION

Purpose and Scope

Utilization of cars has expanded in today's world. The accessible parking slots are not used appropriately. The primary purpose behind this is the inadequate data the client has with respect to accessible parking spots. The driver more often than not takes after a figure based way to deal with discover a parking spot and the majority of the circumstances get disappointed. The driver ends up in an awesome confuse looking a parking spot in his bustling work routine. Numerous a circumstances driver stops in undesired and badly designed spots.

The absence of appropriate office to help him, stop his vehicle prompts to his movement to remote spots for stopping, accordingly bringing on wastage of fuel and clients time. Increasing the number of parking spaces is a solution to this problem but it is not an easily feasible solution because it requires huge investments and is a time- consuming process. Moreover, the presently available parking management system is mostly static and works in a small area. There should be a solution for all the issues of vehicle parking. Use of internet will make system feasible for anyone to use it, as it can be accessible from remote places. So such systems contribute a lot in collecting information for developing a parking management system with a huge database of maps. Everyone in day to day life uses a Smartphone, so the application will be more efficient to the user to operate. This application will help the user to search for parking location without wasting much of time.

OVERVIEW DESCRIPTION

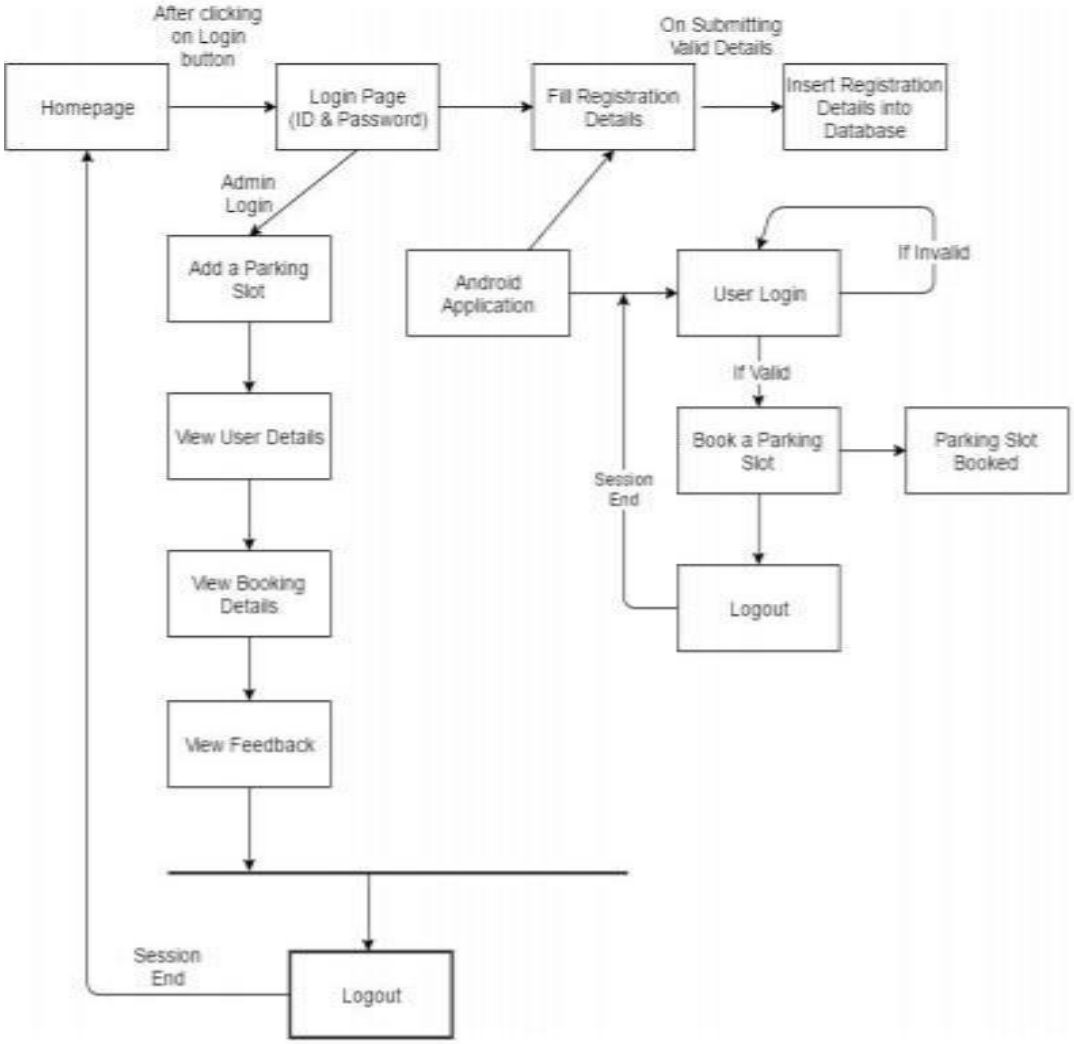
Hardware requirement

Processor speed	1.6GHZ
Memory of user PC	512MB RAM
Disk Space of user PC	5GB
Memory of server Pc	8GB
Bandwidth(network connection)	15Mbps
Disk space of server	200GB

Software Requirements

Operating System for computer	Window 7 windows 8.1,Linux,Mac Os ,Windows 8,windows xp
Database Management System	MySql
Browser	Opera,Google Chrome,IE,Mozilla Firefox

DATA FLOW DIAGRAM



SAMPLE FORMS

ADMIN LOGIN

USER REGISTERATION AND LOGIN FORM

Conclusion:

Thus the Vehicle parking allotment system was implemented using the android application.

EX.NO.: 9	E-TICKETING

PROBLEM STATEMENT

Our project is carried out to develop software for online Railway Reservation System. This system has various options like reservation, cancellation and to view details about available seats. Our project mainly simulates the role of a Railway ticket booking officer, in a computerized way.

SOFTWARE REQUIREMENT SPECIFICATION

PURPOSE

If the entire process of reservation is done in a manual manner then it would takes several months for reservation to reach the applicant. Considering the fact that the number of passenger is increasing every year, an Automated System becomes essential to meet the demand. So this system uses several programming and database techniques to elucidate the work involved in this process. As this is a matter of National Security, the system has been carefully verified and validated in order to satisfy it.

SCOPE

The System provides an online interface to the user where they can fill in their personal details and submit the necessary documents (may be by scanning).

The authority concerned with the issue of railway can use this system to reduce his workload and process the application in a speedy manner.

Provide a communication platform between the passenger and the administrator.

Passenger will come to know their status of application and the date in which they must subject themselves for manual document verification.

OVERALL DESCRIPTION

HARDWARE INTERFACE

The server is directly connected to the client systems. The client systems have access to the database in the server.

SOFTWARE INTERFACE

Front End Client - The passenger and System online interface is built using JSP and HTML.

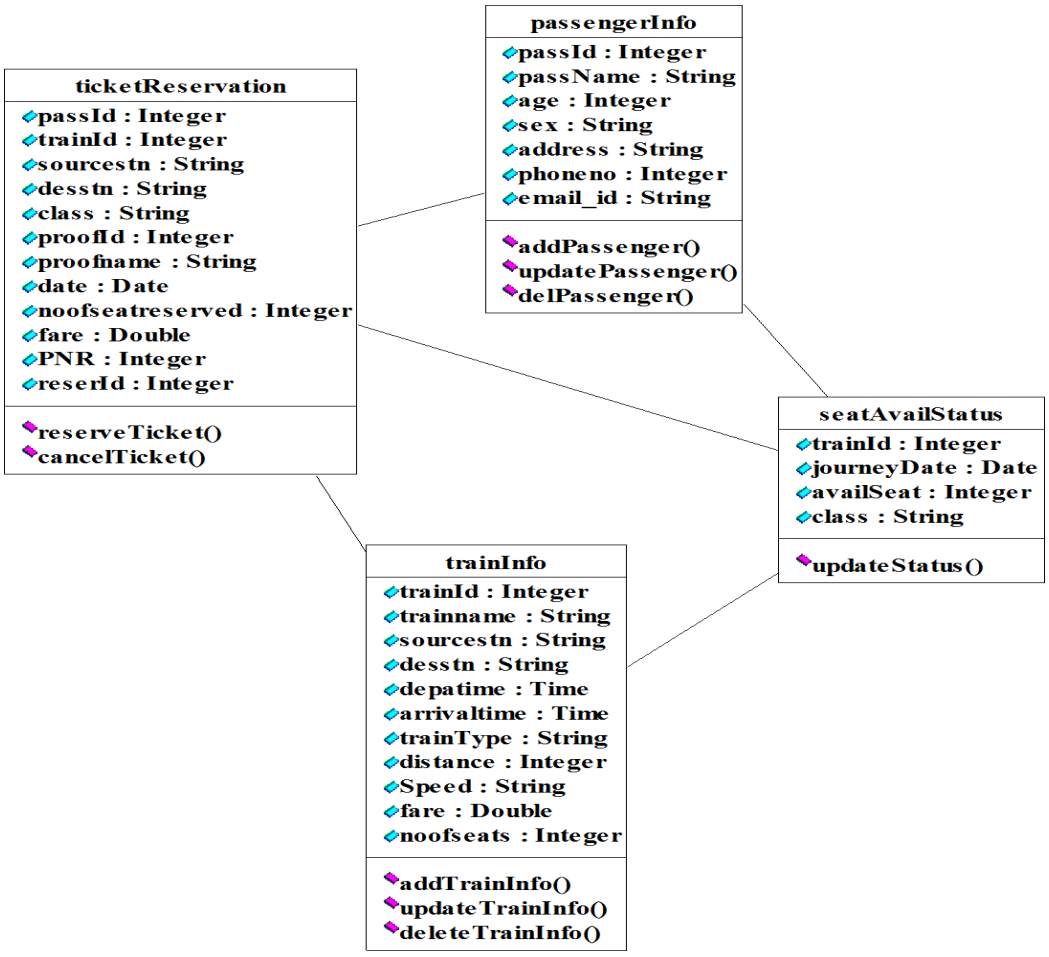
The Administrators's local interface is built using Java.

Web Server – Apache Tomcat Server (Oracle Corporation)

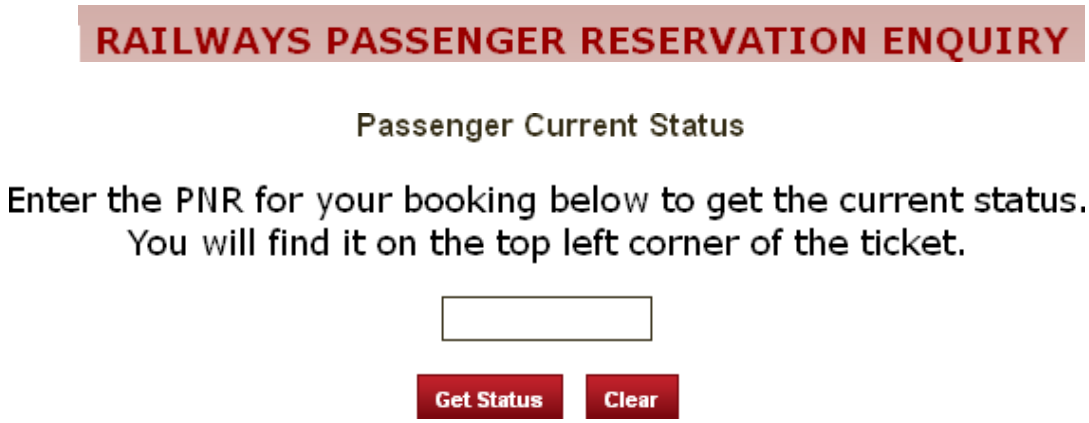
Back End - Oracle 11g database

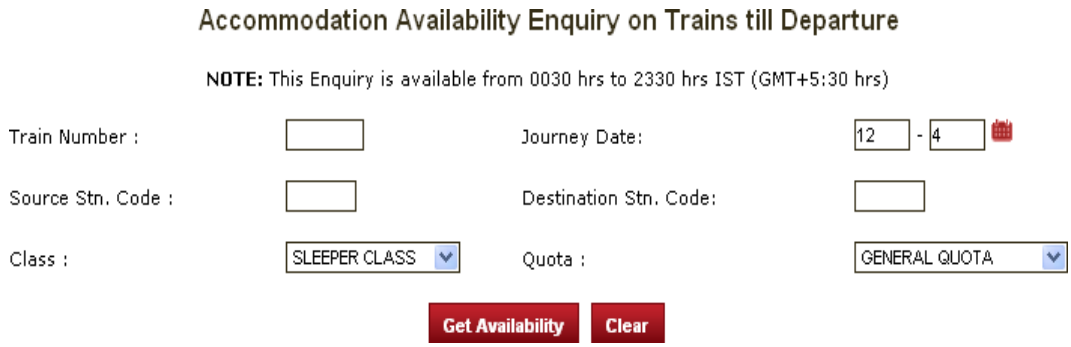
CLASS DIAGRAM:

The class diagram, also referred to as object modeling is the main static analysis diagram. The main task of object modeling is to graphically show what each object will do in the problem domain. The problem domain describes the structure and the relationships among objects.



IMPLEMENTATION OF USER INTERFACE LAYER





Conclusion:

Thus the mini project for railway reservation system has been successfully executed and codes are generated.

EX.NO.: 10	ONLINE COURSE RESERVATION

PROBLEM STATEMENT :

The system is built to be used by students and managed by an administrator. The student and employee have to login to the system before any processing can be done. The student can see the courses available to him/her and register to the course he/she wants. The administrator can maintain the course details and view all the students who have registered to any course.

SOFTWARE REQUIREMENT SPECIFICATION

PURPOSE :

If the entire process of 'Issue of Course' is done in a manual manner then it would takes several months for the course to reach the applicant. Considering the fact that the number of applicants for course is increasing every year, an Automated System becomes essential to meet the demand

SCOPE:

- The System provides an online interface to the user where they can fill in their personal details and submit the necessary documents (may be by scanning).
- The Registrar concerned with the issue of course can use this system to reduce his workload and process the application in a speedy manner.
- Provide a communication platform between the Student and the Registar

OVERALL DESCRIPTION:

HARDWARE INTERFACE:

The server is directly connected to the client systems. The client systems have access
To the database in the server.

SOFTWARE INTERFACE:

Front End Client – The Student and Registrar online interface is built using JSP and

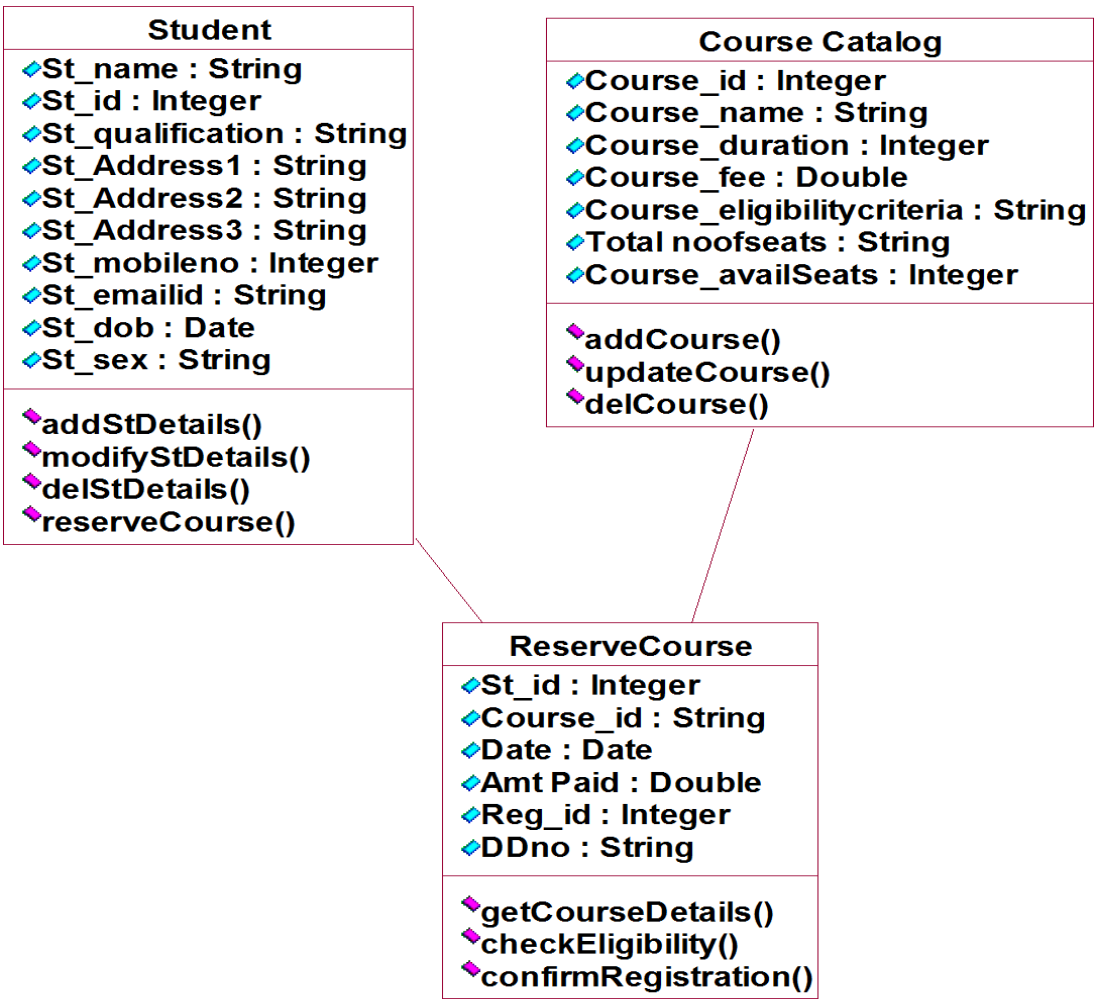
HTML. The Administrators's local interface is
built using Java.

Web Server – Tomcat Apache application server (Oracle Corporation).

Back End – Oracle 11g database.

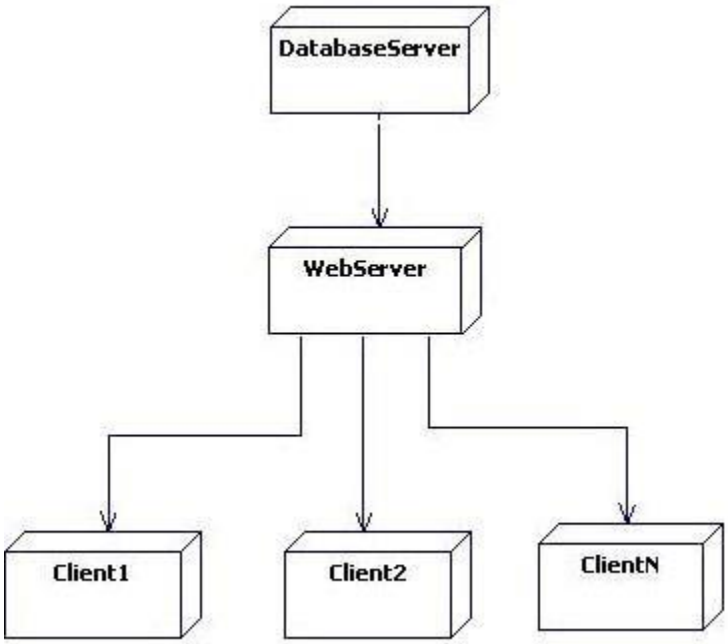
CLASS DIAGRAM:

The class diagram is a graphical representation of all the classes used in the system
And their operations, attributes and relationships.



DEPLOYMENT DIAGRAM AND COMPONENT DIAGRAM

Deployment diagrams are used to visualize the topology of the physical components of a system where the software components are deployed.



Conclusion:

Thus the project for Course Reservation system has been successfully executed and codes are generated.